

Game Jam with CARDinality: A Case Study of Exploring Play-based Interactive Applications

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ABSTRACT

This paper delves into a Game Jam, a workshop-based investigation widespread in the game design community, to explore the potential and applications of CARDinality, a card-shaped robotic device, giving insights into the material implications and prototyping potential of this novel platform. During the Game Jam, 9 participants were informed of basic game design principles and techniques from a game design researcher. They then interacted with our hardware devices to brainstorm future card games supported by CARDinality, revealing a breadth of unique applications anchored in ‘play.’

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices**.

KEYWORDS

Game Jam Workshop, Robotic Device, Tangible Interaction

ACM Reference Format:

Emilie Faracci, Aditya Retnanto, Anup Sathya, Ashlyn Sparrow, and Ken Nakagaki. 2024. Game Jam with CARDinality: A Case Study of Exploring Play-based Interactive Applications. In *The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24)*, October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3672539.3686347>

1 INTRODUCTION

Game Jams are short-term events that focus on the prototyping and rapid development of game ideas [6]. These events, which have gained prevalence in the game design community, have served as a conduit for the exploration of novel game design mediums and technologies. Notable examples include the integration of VR [3] to create immersive game experiences, or the use of ultrasound haptics to generate new game ideas [1]. A recent exploration of ChatGPT in a Game Jam setting [4] provided insights into the capabilities and shortcomings of such LLM tools specific for game design.

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UIST Adjunct '24, October 13–16, 2024, Pittsburgh, PA, USA

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ACM ISBN 979-8-4007-0718-6/24/10

<https://doi.org/10.1145/3672539.3686347>



Figure 1: (a) CARDinality device (b) Game Jam session

In parallel, the Human-Computer Interaction (HCI) community has explored the development of novel hardware technology to envision future materials that can sense [9] and actuate [5, 7]. As computation and interactive capabilities fuse into future material design, it is important for HCI researchers to explore how such technologies would be employed and inspire future designers in downstream domains.

Given this context, Game Jams stand out for their potential to uncover new ideas and applications in interaction design with emerging robotic technologies. In this paper, we explore a Game Jam with the new robotic material medium of CARDinality [8] (Figure 1a), a system of flat, card-shaped robotic devices that use vibration for locomotion and haptics. As a case study anchored in a game design workshop employing novel robotic devices, we contribute to the HCI community by providing (A) Characteristics of Game Jam ideation for revealing the potential of actuated tangible technology and (B) How CARDinality and materials with computation, in general, can enhance gameplay, expanding the possibilities of the device.

Overview of CARDinality: CARDinality [8] hardware features a thin and card-like form factor (employing thin, flexible PCB and battery) and enables wireless communication. The devices contain several primary modalities for interaction: **IMU sensing** of card status, and gesture, on-table **locomotion** and auditory **Buzz-Notification**, and in-hand **haptics**.

2 GAME JAM

2.1 Procedure

The Game Jam (Figure 1b) started with an introduction to game design mechanics and affordances of new game design mediums,

		# ideas per group		Main Functionalities Used						
		Team 1	Team 2	Sensing			Locomotion		Actuation	
Game Category	Example Interaction / Existing Game (tagged with *)			State	Gesture	Surface			Haptics	Buzz
Autonomous Characters	Pokemon*, Fantasy Football*, Tamagotchi*, Tag, Race, Settlers of Catan*	6	10			✓		✓		
Rhythm & Dexterity Games	Music Tapping Game, Plane Controller, House of Cards, Hot Potato*, Domimos*	3	2	✓	✓					
Novel Communication & Interactions	Morse Code, Secret Vote, Vibration as Music	3	5		✓				✓	✓
Proximity and Movement	Hide and Seek*, Marco Polo*, Hot and Cold*, Moving figurines	1	4	✓				✓		
Computational & Educational Tool	Time Keeper, Game Rule Explainer, Card Counting Tool	1	1						✓	✓
Social Deduction Games	Mafia*, Werewolf*, Kent*, Guess Who*	1	3		✓			✓	✓	
Other Traditional Games	Bridge*, Uno*	0	1						✓	

Figure 2: Idea analysis chart from the initial brainstorming session, representing the breadth of ideas given by the participants, and functionalities of the device each idea of game category employed.

including methods and concepts from *Building Blocks of Tabletop Game Design* [2]. One of our co-authors had extensive experience in game design theory and Game Jam organization to lead this course.

Then, the participants were introduced to the capabilities of the device and a Graphical User Interface (GUI) to test its input/out (I/O) functionality and feel it in their hands. There were two rounds of ideation: **1. open-ended brainstorming session using post-it (1 hour)** to diverge the ideas, and **2. polished a single idea (30 mins)** in a team of 4 or 5, culminating in a presentation of their fully-fledged game. We prioritized ideas not to be constrained by control software and intended to reveal the future potential of the device, so the final ideas didn't require demonstration with the hardware. The study was reviewed and approved by our institution's ethics committee (protocol IRB24-0548).

Participants and Apparatus: The Game Jam included 9 participants (aged 19-27 years old; 4 self-identified as females, 5 as males) from our institution with some degree of familiarity in game design, hardware prototyping, or programming. There were 8 devices shared over two desktop computers to wirelessly control them.

2.2 Results

Brainstorming Ideas: Participants explored a wide variety of ideas, as displayed in Figure 2. They devised games where CARDinality cards could represent *Autonomous Characters*, using locomotion for games like racing, tag, or maze solving. The card form-factor also allowed for *Rhythm and Dexterity Games* like using CARDinality as a plane controller or making a house of cards. Participants tested different sensing modalities, such as tapping the cards, for *Novel Communication and Interactions*. By using sensing and the mobility of cards, participants also came up with *Proximity and Movement* examples. The computational capabilities of CARDinality were used for a variety of *Computational and Educational Tools*, like a time keeper or game rule explainer. The functionalities of CARDinality proved especially compatible with *Social Deduction Games* like Werewolf¹. Finally, *Traditional Games* like Bridge were reimaged to incorporate CARDinality for innovative communication mechanics, using locomotion to move to players, or haptics to communicate with the entire table.

Final Ideas: Team 1 devised a variation of the game *Kent*², incorporating CARDinality's features to introduce new dynamics. Players

could communicate using card tapping, with specific patterns indicating special power cards or actions. These cards, distributed randomly throughout the deck, added unpredictability and strategy to the game.

Team 2 proposed '**Magic Uno**,' a variation of the classic Uno³ game enhanced by CARDinality's capabilities. Special cards could vibrate, signaling the activation of their powers, and players had to use them strategically or risk losing their abilities, introducing an element of suspense and randomness to the game.

2.3 Findings

The findings from this Game Jam are two-fold. First, with regards to **characteristics of Game Jam ideation for revealing the potential for new technology**: Game Jams prove to be a playful method to brainstorm, nudging creativity over practicality. Because of this ethos and emphasis on prototyping over a finished product, they can allow for a breadth of ideas, broadening the range of possibilities for new technology. People can also anchor ideas with existing games, which they often collectively know, and this becomes a catalyst to ideate interaction design concepts.

Second, for **How CARDinality or materials with computation can enhance game play**: in this workshop, we noticed that sensing became a catalyst for actuation, haptics was mainly used for communication, and locomotion was used to create novel autonomous behavior. Participants used the Game Jam to highlight how computation can augment the play, through unpredictability and randomization, for example.

3 CONCLUSION AND FUTURE WORK

This hands-on Game Jam opened up exciting avenues for future application in game and play with CARDinality, providing insights into the use of this novel game design medium. While the Game Jam presented here focused on the game ideas and concepts themselves, future research should facilitate participants' ability to build and prototype the ideas with the hardware devices, in which case users could test-play their games and advance their ideas through collaboration and iteration.

ACKNOWLEDGMENTS

We appreciate Actuated Experience Lab members for the help with the project.

¹<https://playwerewolf.co/>

²[https://en.wikipedia.org/wiki/Kempes_\(card_game\)](https://en.wikipedia.org/wiki/Kempes_(card_game))

³<https://www.letsplayuno.com/>

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